



PUZZLE ITC
changing IT for the better

Backstage Workshop

Christoph Raaflaub
raaflaub@puzzle.ch

Agenda

- Introduction
- Platform Engineering
- Developer Portal
- Backstage
 - Architecture
 - Plugin Framework
 - System Model
- Hands-on



Nice to meet you



Christoph Raaflaub
Platform Architect
Member of the Technical Board
raaflaub@puzzle.ch

Introduction Round

- Introduce yourself
- Knowledge / Experience
 - Platform Engineering
 - Internal Developer Platform
 - Developer Portal (like Backstage)
- Expectations for the workshop



Platform Engineering

Why Platform
Engineering

Why Platform Engineering?

CLOUD NATIVE LANDSCAPE [EXPLORE](#) [GUIDE](#) [STATS](#)

Application Definition and Image Build [Application Definition and Image Build](#) [Streaming & Messaging](#)

Continuous Integration & Delivery [Continuous Integration & Delivery](#) [Database](#)

Scheduling & Orchestration [Scheduling & Orchestration](#) [Service Proxy](#) [API Gateway](#)

Service Mesh [Service Mesh](#) [Coordination & Service Discovery](#) [Remote Procedure Call](#)

Container Runtime [Container Runtime](#) [Cloud Native Storage](#) [Cloud Native Network](#)

Security & Compliance [Security & Compliance](#) [Container Registry](#)

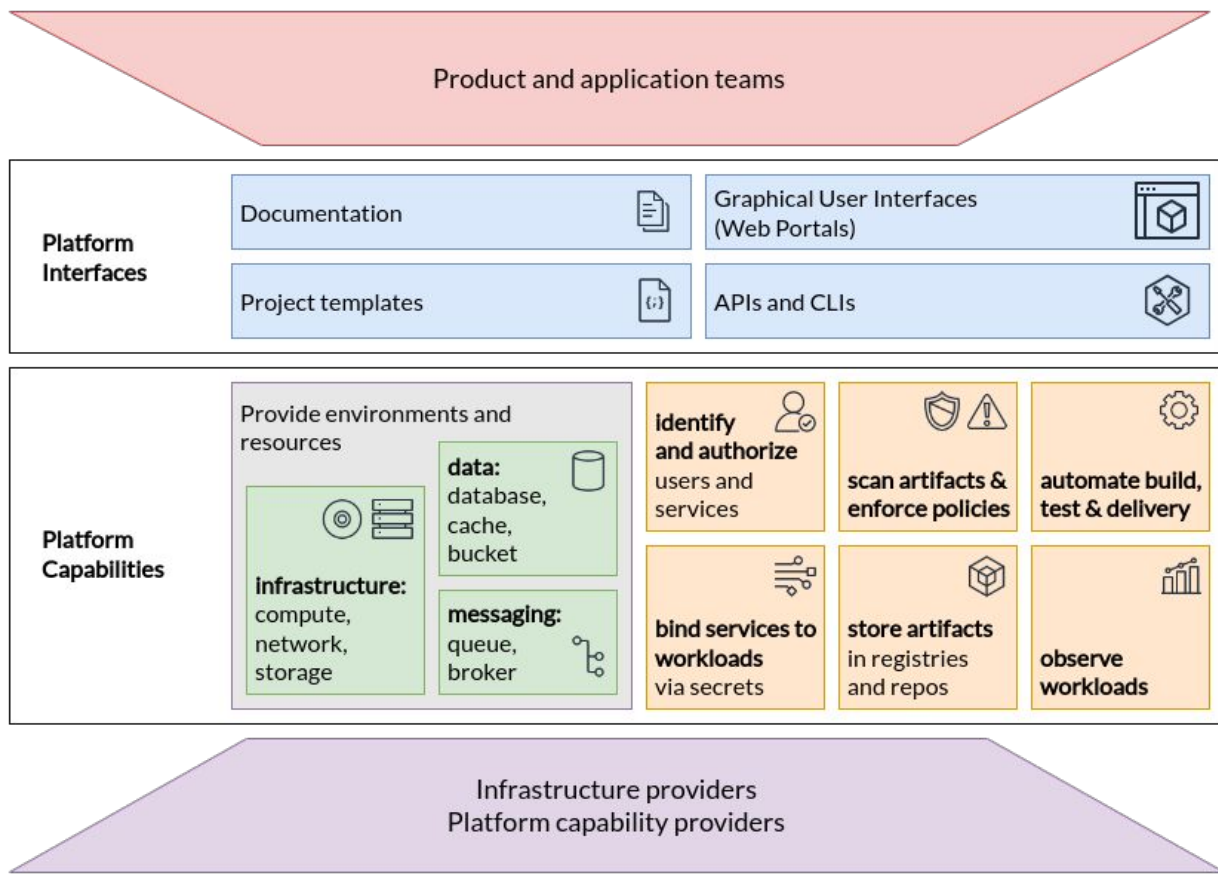
Automation & Configuration [Automation & Configuration](#) [Key Management](#)

Observability and Analytics [Observability and Analytics](#) [Feature Flagging](#)

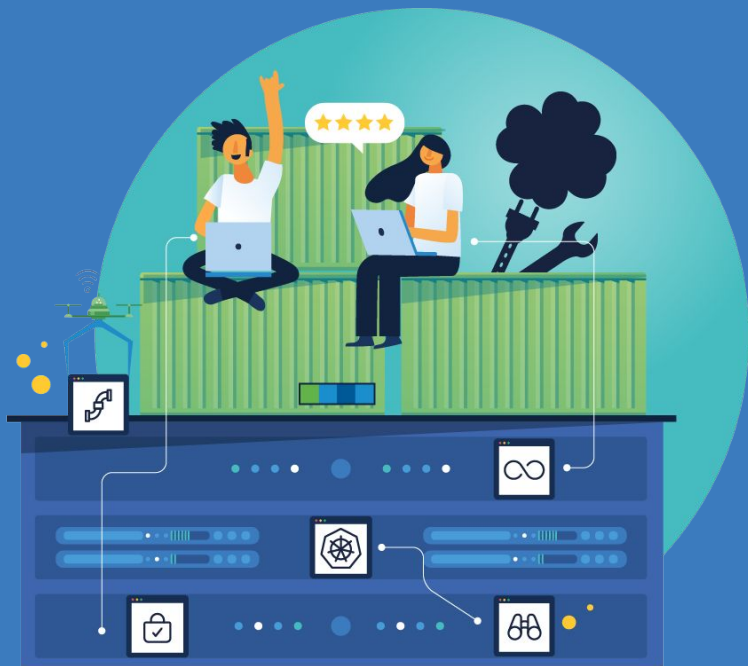
Chaos Engineering [Chaos Engineering](#) [Continuous Optimization](#)

<https://landscape.cncf.io/>

What is a Platform?



Platform Engineering



"Why we should stop
'tinkering' with
infrastructure and start
building it as a product."

Platform Engineering

- Platform as a Product
- Reduce Cognitive Load
- Self-Service
- Golden Paths

Internal Developer Platform vs. Portal



Platform

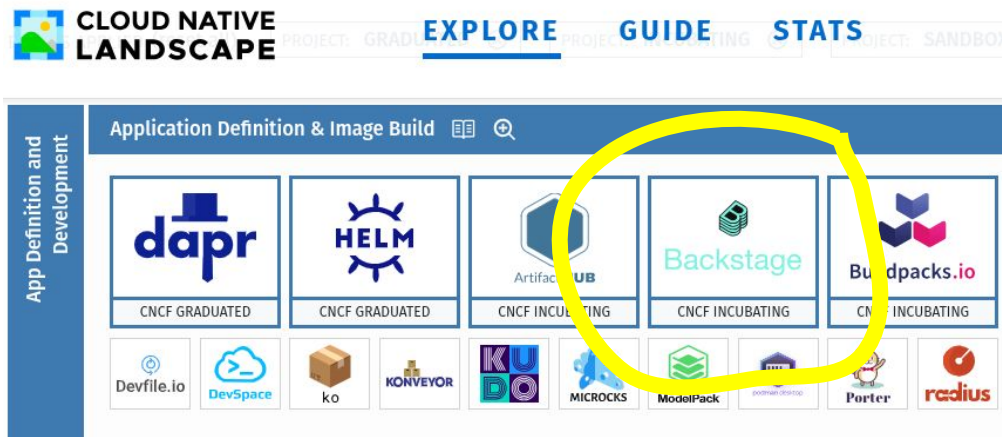
The underlying APIs and tools that provide resources



Portal

The user interface (GUI) that bundles and visualizes the platform complexity

Backstage



Backstage is an **open platform** for building developer portals.

- Developed by Spotify
- Open sourced in March 2020
- CNCf [Incubation](https://landscape.cncf.io/) Project
- Plugin System

Backstage Portal



Service Catalog

Single Source of Truth for all software components



Software Templates

Create a new service at the touch of a button (Golden Paths).



Single Pane of Glass

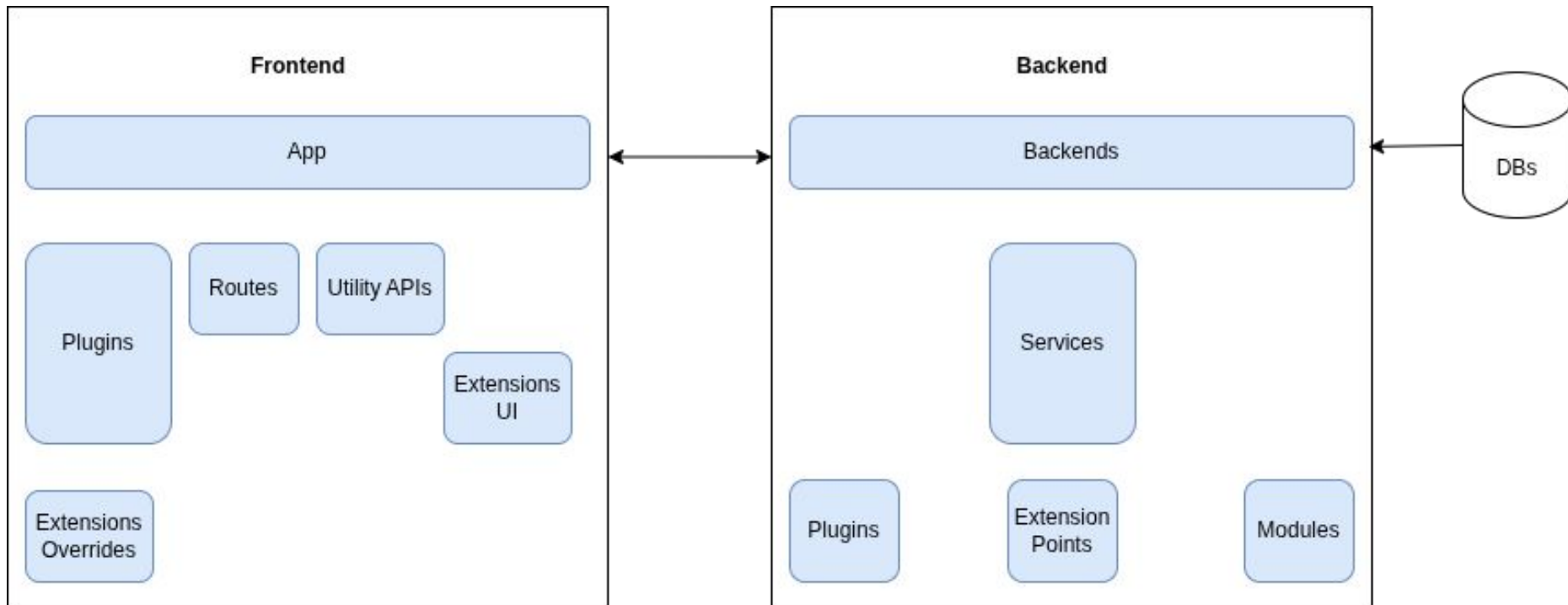
Aggregates the most important information in one place.



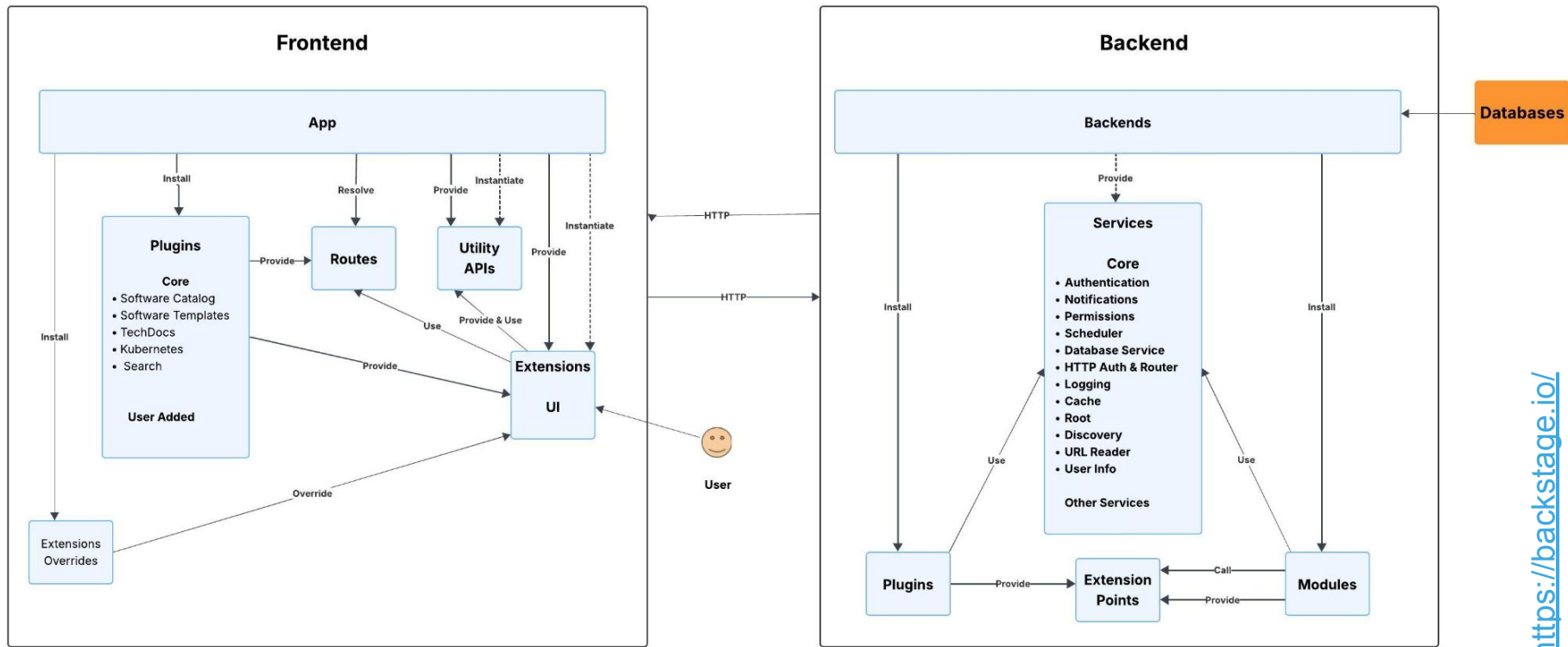
Self-Service

Infrastructure requirements can be triggered directly via the portal, instead of writing tickets.

Backstage Architecture



Backstage Architecture



It is your turn, get your hands dirty

Hands-on guide:

<https://backstage-techlab.puzzle.ch/>

